

Qualification Pack



Technician - Tower Construction

Electives: Erection and Stringing/ Foundation

QP Code: PSS/Q3501

Version: 1.0

NSQF Level: 4

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PSS/Q3501: Technician - Tower Construction

Brief Job Description

The individual in this job carries out erection, installation and commissioning of power transmission towers of all types.

Personal Attributes

The individual must be physically and mentally able to perform essential functions of the job safely. Work is performed mostly outdoors in all weather conditions. He/she will be required to perform manual activities for long periods of time.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1331: Apply basic health and safety practices for power related work](#)
2. [PSS/N1336: Working effectively with others](#)

Electives (mandatory to select at least one):

Elective 1: Erection and Stringing

Specializes in Tower Erection and Stringing of Transmission lines.

1. [PSS/N3501: Perform erection and stringing of a transmission tower](#)

Elective 2: Foundation

Specializes in Foundation of Transmission Tower

1. [PSS/N3502: Construct the foundation of a transmission tower](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Transmission



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Occupation	Erection, Installation and Commissioning - Transmission
Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	10th Class
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	11/03/2019
Next Review Date	11/03/2023
NSQF Approval Date	22/08/2019
Version	1.0
Reference code on NQR	2019/POW/PSSC/03482
NQR Version	1.0



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PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Scope

This unit/task covers the following:

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** • use protective clothing/equipment for specific tasks and work conditions
Protective Clothing
• leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visor
equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace
Hazards- electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards
possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions
- PC3.** follow procedures for safe working with electrical equipment
Safe Working Procedures: tag out/lock out, ptw (permit to work)
- PC4.** follow warning signs while working with electrical systems
Warning Signs: danger, high voltage area, out of service, etc
- PC5.** use standard safe working practices when working at heights, confined areas and trenches
- PC6.** test any electrical equipment and system using insulated testing devices before touching them
- PC7.** ensure positive isolation of electrical equipment & system as per the given standards
- PC8.** identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installed
Parameters: temperature, pressure, flow and current

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- PC9.** follow safe working practices while working in hazardous conditions to ensure the safety of self and others
Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.
- PC10.** use various methods of accident prevention in the work environment
Methods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors
- PC11.** identify and retrieve general health and safety equipment in the workplace
General Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.
- PC12.** set up and use scaffolds, elevated platforms and ladders safely
Set up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc
- PC13.** inspect scaffolds, elevated platforms and ladders for faults and rectify them
Faults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.
- PC14.** lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa
- PC15.** inspect power plant and its equipment routinely for any signs of oil, water and steam leakage
- PC16.** store flammable materials and machine lubricating oil safely and correctly
- PC17.** check that the emission and pollution control devices are working properly in line with environmental policy standards
- PC18.** use good housekeeping practices at all times
Good Housekeeping Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces
- PC19.** identify common hazard signs displayed in various areas
Various Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.
- PC20.** inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

- PC21.** use various appropriate fire extinguishers for different types of fires correctly
Types of Fires: class a, b, c, d and e
- PC22.** use appropriate rescue techniques during fire hazard
- PC23.** maintain good housekeeping in order to prevent fire hazards
- PC24.** use a fire extinguisher correctly in case of fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC25.** administer appropriate first aid to the victims for the relevant injuries
Injuries: bleeding, burns, choking, electric shock, poisoning etc.
- PC26.** act promptly and appropriately to an accident situation or medical emergency in real or simulated environments



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- PC27.** perform and organise loss minimization or rescue activity during an accident in real or simulated environments
- PC28.** administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services
- PC29.** demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergencies
Emergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work
- PC30.** free an affected person from electrocution safely as per the recommended procedure
- PC31.** move injured people and others to a safe place during emergencies
- PC32.** write an incident report or dictate a report to another person, and send report to the authorized person
Incident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. names (and job titles if applicable), and where to find, all the people responsible for health and safety in a workplace
2. names and location of documents that refer to health and safety in the workplace
3. names and location of people responsible for health and safety in the workplace
4. documents that refer to health and safety in the workplace
documents: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (e.g. government notices)
5. meaning of hazards and risks
6. health and safety hazards commonly present in the work environment and related precautions
7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
8. possible causes of risk and accident
possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness); not taking safety precautions
9. methods of accident prevention
10. safe working practices when working with tools and machines
11. safe working practices while working at various hazardous sites
12. where to find all the general health and safety equipment in the workplace
13. various dangers associated with the use of electrical equipment
14. positive isolation of electrical equipment and system
15. safe handling and disposal of hazardous power plant wastes
16. use of emission and pollution control devices and measures taken to control pollution
17. various safety procedures and equipment used to work at heights, trenches and confined places
18. safe working practices specific to working with electrical equipment and system e.g. lock out/tag out, ptw, etc.



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19. preventative and remedial actions to be taken in the case of exposure to toxic materialstoxic materials: solvents, flux, lead exposure: ingested, contact with skin, inhaledpreventative action: ventilation, masks, protective clothing/ equipment); remedial action: immediate first aid, report to supervisor
20. importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
21. precautionary measures taken to prevent fire accident
22. various causes of firecauses of fires: heating of metal; spontaneous ignition; sparking; electrical heating; loose fires (smoking, welding, etc.); chemical fires; etc.
23. correct technique(s) of using a fire extinguisher
24. different methods of extinguishing fire
25. different materials used for extinguishing firematerials: sand, water, foam, co2, dry powder
26. emergency rescue techniques applied during a fire hazard
27. various types of safety signs and what they mean
28. appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
29. rescue techniques applied during fire hazards
30. good housekeeping in order to prevent fire hazards
31. correct use of a fire extinguisher
32. how to free a person from electrocution
33. basic techniques of bandaging
34. perform artificial respiration and the cpr process

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write an accident/ incident report in local language and/or english
2. read and comprehend basic content to read labels, charts, signages
3. read and comprehend basic english to read manuals of operation
4. read an accident/ incident report in local language or english
5. question coworkers appropriately in order to clarify instructions and other issues
6. give clear instructions to coworkers and subordinates
7. remain congenial while discussing and debating issues with co-workers
8. follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
9. ask for, provide and receive required assistance where possible to ensure achievement of work-related objectives
10. follow organizational rule-based decision-making process
11. take decisions with systematic course of actions and/or response
12. plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
13. build customer relationships and use customer centric approach



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14. think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
15. seek appropriate assistance from other sources to resolve problems
16. report problems that cannot be resolved to the appropriate authority
17. identify cause and effect relations in own area of work
18. work systematically and logically to resolve the issues, identify the cause and anticipate unexpected results
19. critically evaluate the organizational processes to improve them

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	19	43	-	-
PC1. • use protective clothing/equipment for specific tasks and work conditions Protective Clothing • leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visor equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator	1	2	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace Hazards-electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions	1	2	-	-
PC3. follow procedures for safe working with electrical equipment Safe Working Procedures: tag out/lock out, ptw (permit to work)	1	2	-	-
PC4. follow warning signs while working with electrical systems Warning Signs: danger, high voltage area, out of service, etc	1	2	-	-
PC5. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC6. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC7. ensure positive isolation of electrical equipment & system as per the given standards	1	1	-	-
PC8. identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installed Parameters: temperature, pressure, flow and current	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. follow safe working practices while working in hazardous conditions to ensure the safety of self and others Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.	1	2	-	-
PC10. use various methods of accident prevention in the work environment Methods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors	1	2	-	-
PC11. identify and retrieve general health and safety equipment in the workplace General Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.	1	2	-	-
PC12. set up and use scaffolds, elevated platforms and ladders safely Set up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc	1	3	-	-
PC13. inspect scaffolds, elevated platforms and ladders for faults and rectify them Faults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/unfixed nuts or bolts, etc.	1	3	-	-
PC14. lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa	-	3	-	-
PC15. inspect power plant and its equipment routinely for any signs of oil, water and steam leakage	1	3	-	-
PC16. store flammable materials and machine lubricating oil safely and correctly	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	2	-	-
PC18. use good housekeeping practices at all times Good Housekeeping Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces	1	2	-	-
PC19. identify common hazard signs displayed in various areas Various Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.	1	2	-	-
PC20. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	2	-	-
<i>Follow fire safety practices</i>	3	8	-	-
PC21. use various appropriate fire extinguishers for different types of fires correctly Types of Fires: class a, b, c, d and e	1	2	-	-
PC22. use appropriate rescue techniques during fire hazard	1	2	-	-
PC23. maintain good housekeeping in order to prevent fire hazards	1	2	-	-
PC24. use a fire extinguisher correctly in case of fire	-	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	19	-	-
PC25. administer appropriate first aid to the victims for the relevant injuries Injuries: bleeding, burns, choking, electric shock, poisoning etc.	1	2	-	-
PC26. act promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC27. perform and organise loss minimization or rescue activity during an accident in real or simulated environments	1	4	-	-
PC28. administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergencies Emergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC30. free an affected person from electrocution safely as per the recommended procedure	1	2	-	-
PC31. move injured people and others to a safe place during emergencies	-	3	-	-
PC32. write an incident report or dictate a report to another person, and send report to the authorized person Incident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified	2	2	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Distribution, Transmission, Generation, Distribution Downstream
Occupation	Operation & Maintenance - Generation Supervision - Distribution Erection, Installation and Commissioning - Distribution Surveying - Transmission Operation & Maintenance - Transmission
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	11/02/2023
NSQC Clearance Date	22/08/2019



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PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU6.** importance of effective communication in the workplace
- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication



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- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge
- GS2.** note down observations (if any) related to the operation/maintenance
- GS3.** read and interpret the process required for different types of manuals
- GS4.** read and interpret the flowchart of all parts of an assembly
- GS5.** read manuals and documents to understand the product-details & how they can be used
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor
- GS7.** effectively communicate with the team members
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor
- GS9.** communicate clearly with the colleague on the issues faced during query/fault
- GS10.** follow colleague/contractor rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response
- GS12.** planning and organization of tasks to meet deadlines
- GS13.** build customer relationships and use customer centric approach
- GS14.** seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
- GS15.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
- GS16.** critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>working with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023



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PSS/N3501: Perform erection and stringing of a transmission tower

Description

This unit covers the erection of a transmission tower and stringing of a transmission line as per procedures.

Scope

This unit/ task covers the following:

- Perform erection of a transmission tower as per specifications and standard operating procedures
- Perform stringing of a transmission line as per specifications and standard operating procedures

Elements and Performance Criteria

Perform erection of a transmission tower as per specifications and standard operating procedures

To be competent, the user/individual on the job must be able to:

- PC1.** read and interpret the instructions for erection of the transmission tower
- PC2.** identify various tools and machineries required for tower erection
- PC3.** receive and check the materials, fasteners, members etc. of tower
- PC4.** ensure proper loading, unloading and storage of the materials and components at the site
- PC5.** check that laying of the foundation including stub setting is as per the layout drawing
- PC6.** carry out section-wise ground assembly of the transmission tower
- PC7.** carry out proper fitting and arrangement of sections, cross arm and peck assembly as per instructions given by the concerned authority
- PC8.** carry out tightening of fasteners and punching of threads as per requirement
- PC9.** perform tack welding of nuts of the transmission tower
- PC10.** paint joints of the transmission tower
- PC11.** carry out earthing of the tower

Perform stringing of a transmission line as per specifications and standard operating procedures

To be competent, the user/individual on the job must be able to:

- PC12.** mount insulators
- PC13.** lay earth wire and conductors
- PC14.** check the tension and ground clearance of earth wire and conductors
- PC15.** fix other accessories such as anti-climbing device, danger plates, bird guard, dampers etc. as per requirement
- PC16.** run conductor/rope through travelers to reduce friction in case of sagging
- PC17.** attach pulling equipment to conductor/ rope
- PC18.** perform paving of conductors on the ground along the tower taking into account permissible span length and sag
- PC19.** transfer the conductor from traveler to insulator
- PC20.** secure the conductor using clamps and ties



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- PC21.**
- perform post
 - installation procedures for ensuring clean and safe environment in the work and surrounding area

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the organization relevant to own employment and performance conditions
2. relevant health and safety requirements applicable in the work place
3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
4. reporting structure, inter-dependent functions, lines and procedures in the work area
5. how to engage with approved specialists for support in order to resolve incidents and service requests
6. relevant people and their responsibilities within the work area
7. escalation matrix and procedures for reporting work and employment related issues
8. common electricity terminology: current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)
9. correct use of various tooltools: spanners, hammer, chisels, gauges and other basic tools required in a power plant
10. types of transmission tower based on height, transmission voltage and technology
11. various types of erection, installation and commissioning methods
12. material requirements for erection and commissioning
13. tower erection process
14. tools and tackles used in tower erection
15. how to access and use technical manuals, blueprints, schematics, diagrams, plans, specifications
16. ht transmission system and its components
17. various stringing methods
18. procedure to be used for stringing of conductors
19. relevance of sag in a transmission line
20. importance of following procedures to achieve optimum sag- tension
21. problems that can occur with tower erection and how these can be overcome
22. methods and parameters to check quality of the components against required quality standards
23. recording procedures including data logging, chart recording of various activities and data points like tolerance level etc.
24. how to estimate time, material and equipment needed to complete assignments
25. how to tighten nuts and punch threads
26. fundamentals of tack welding
27. process of painting of joints of transmission tower
28. process of checking the verticality of erected transmission towers



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29. how to use/operate tower erection tools such as spanners, pulley arrangement, rope, 2-way, 3-way pulleys, nuts, bolts, tack welding equipment, step bolts etc.
30. process of handling all machineries, equipment and vehicles for tower erection
31. quality parameters, quality assessment processes based on physical parameters
32. importance of reporting problems in a timely manner
33. importance of working in a clean environment, safe working practices and procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

1. note the information communicated by the supervisor or engineer
2. note down observations (if any) related to the process
3. read and correctly interpret the process required for various types of operations
4. read and correctly interpret the process flowchart required for various types of operations
5. read manuals and documents related to the tasks to identify the tasks and interpret the equipment functioning
6. communicate effectively with team and other staff members
7. discuss task lists, schedules and activities with colleagues
8. follow organization rule-based decision-making process
9. plan and organize tasks to meet targets
10. seek and comprehend information required for effective problem solving
11. apply domain information to set and define operation parameters that ensure economy and quality of the service
12. critically evaluate operation parameters with intended targets in order to ensure a match



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform erection of a transmission tower as per specifications and standard operating procedures</i>	13	39	-	-
PC1. read and interpret the instructions for erection of the transmission tower	1	2	-	-
PC2. identify various tools and machineries required for tower erection	2	4	-	-
PC3. receive and check the materials, fasteners, members etc. of tower	1	4	-	-
PC4. ensure proper loading, unloading and storage of the materials and components at the site	1	4	-	-
PC5. check that laying of the foundation including stub setting is as per the layout drawing	2	5	-	-
PC6. carry out section-wise ground assembly of the transmission tower	1	4	-	-
PC7. carry out proper fitting and arrangement of sections, cross arm and peck assembly as per instructions given by the concerned authority	1	5	-	-
PC8. carry out tightening of fasteners and punching of threads as per requirement	1	2	-	-
PC9. perform tack welding of nuts of the transmission tower	1	3	-	-
PC10. paint joints of the transmission tower	1	2	-	-
PC11. carry out earthing of the tower	1	4	-	-
<i>Perform stringing of a transmission line as per specifications and standard operating procedures</i>	12	36	-	-
PC12. mount insulators	1	4	-	-
PC13. lay earth wire and conductors	1	5	-	-
PC14. check the tension and ground clearance of earth wire and conductors	1	5	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. fix other accessories such as anti-climbing device, danger plates, bird guard, dampers etc. as per requirement	1	4	-	-
PC16. run conductor/rope through travelers to reduce friction in case of sagging	2	4	-	-
PC17. attach pulling equipment to conductor/rope	1	3	-	-
PC18. perform paving of conductors on the ground along the tower taking into account permissible span length and sag	1	3	-	-
PC19. transfer the conductor from traveler to insulator	2	3	-	-
PC20. secure the conductor using clamps and ties	1	2	-	-
PC21. - perform post - installation procedures for ensuring clean and safe environment in the work and surrounding area	1	3	-	-
NOS Total	25	75	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N3501
NOS Name	Perform erection and stringing of a transmission tower
Sector	Power
Sub-Sector	Transmission
Occupation	Erection, Installation and Commissioning - Transmission
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	11/03/2019
Next Review Date	11/03/2023
NSQC Clearance Date	22/08/2019



Qualification Pack

PSS/N3502: Construct the foundation of a transmission tower

Description

This unit covers the preparation and formation of a transmission tower foundation as per procedures.

Scope

This unit/ task covers the following:

- Prepare for the foundation construction process
- Carry out foundation construction process

Elements and Performance Criteria

Prepare for the foundation construction process

To be competent, the user/individual on the job must be able to:

- PC1.** read and interpret drawings/diagrams of transmission tower foundation with respect to different types of tower
- PC2.** identify tools required for excavation and foundation work including rcc (reinforced cement concrete)
- PC3.** issue an indent for machinery required for activities related to tower foundation
- PC4.** perform pre-foundation stage activities such as survey, tower spotting as per drawing, soil investigation as per standards etc.

Carry out foundation construction process

To be competent, the user/individual on the job must be able to:

- PC5.** check the excavation process and ensure the work is as per design parameters
- PC6.** perform reinforcement work (steel bar and binding work) as per standards
- PC7.** perform shuttering of foundation
- PC8.** prepare and pour the concrete mixture as per specification followed by compaction
- PC9.** align and level concrete stubs/ embedded parts
- PC10.** provide provision for earthing the equipment
- PC11.** coordinate with the concerned personnel to conduct earth resistance testing
- PC12.** remove the shuttering after performing the curing process
- PC13.** check foundation for any voids or cracks with approved method
- PC14.** ensure backfilling of earth following standard procedure
- PC15.** inspect the tower foundations and stubs etc. for any damage and perform its rectification/ reinforcement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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1. relevant legislation, standards, policies, and procedures followed in the organization relevant to own employment and performance conditions
2. relevant health and safety requirements applicable in the work place
3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
4. reporting structure, inter-dependent functions, lines and procedures in the work area
5. how to engage with specialists for support in order to resolve incidents and service requests
6. relevant people and their responsibilities within the work area
7. common electricity terminology: current, voltage, resistance, kilowatt (kw), kilowatt hour(kwh)
8. correct use of various tools: spanners, hammer, chisels, gauges and other basic tools required in a power plant
9. various types of transmission tower
10. various type of tower foundation based on type of transmission tower
11. types of foundation, stub setting, material testing etc.
12. strata of land and geography transmission line area
13. types/components/members of tower
14. constituents of foundation material
15. tower foundation process
16. tools and tackles used in tower foundation
17. technical manuals, blueprints, schematics, diagrams, plans, specifications
18. estimate time, material and equipment needed to complete assignments
19. handling of all machineries, equipment and vehicles related to tower foundation
20. use of appropriate judgment and initiative pertaining to work methods and tools
21. guidelines for ground clearance of stubs
22. quality parameters, quality assessment based on physical parameters
23. importance of working in a clean environment and safe practices and procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write official notes
2. write the information communicated by the work in-charge
3. write about the technical problems and other conditions of the site as per standards
4. note critical points and testing repair observations
5. write about the condition of equipment
6. fill in all technical forms and data as per guidelines and format
7. read and correctly interpret the process required for various types of operations
8. read and correctly interpret the process flowchart for all operations
9. read and correctly interpret equipment manuals to identify operational details and specifications



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10. communicate clearly with team and other staff members
11. discuss task lists, schedules and activities
12. interact with the junior engineer to clarify instructions and doubts
13. make appropriate work related judgments based on policy and procedures
14. identify complex problems and review related information to develop and evaluate solutions
15. follow organizations rule-based decision making process
16. plan and organize tasks to meet quantity and quality specifications and targets
17. seek and comprehend information required for effective problem solving
18. apply domain information to set and define operation parameters that ensure economy and quality of the service
19. critically evaluate operation parameters in relation to the intended targets to ensure a match

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for the foundation construction process</i>	9	19	-	-
PC1. read and interpret drawings/diagrams of transmission tower foundation with respect to different types of tower	2	4	-	-
PC2. identify tools required for excavation and foundation work including rcc (reinforced cement concrete)	2	4	-	-
PC3. issue an indent for machinery required for activities related to tower foundation	2	4	-	-
PC4. perform pre-foundation stage activities such as survey, tower spotting as per drawing, soil investigation as per is standards etc.	3	7	-	-
<i>Carry out foundation construction process</i>	21	51	-	-
PC5. check the excavation process and ensure the work is as per design parameters	3	3	-	-
PC6. perform reinforcement work (steel bar and binding work) as per standards	2	6	-	-
PC7. perform shuttering of foundation	2	7	-	-
PC8. prepare and pour the concrete mixture as per specification followed by compaction	2	7	-	-
PC9. align and level concrete stubs/ embedded parts	2	4	-	-
PC10. provide provision for earthing the equipment	2	3	-	-
PC11. coordinate with the concerned personnel to conduct earth resistance testing	2	4	-	-
PC12. remove the shuttering after performing the curing process	-	4	-	-
PC13. check foundation for any voids or cracks with approved method	2	4	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. ensure backfilling of earth following standard procedure	1	4	-	-
PC15. inspect the tower foundations and stubs etc. for any damage and perform its rectification/ reinforcement	3	5	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N3502
NOS Name	Construct the foundation of a transmission tower
Sector	Power
Sub-Sector	Transmission
Occupation	Erection, Installation and Commissioning -Transmission
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	11/03/2019
Next Review Date	11/03/2023
NSQC Clearance Date	22/08/2019

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- 1.Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70



Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1331.Apply basic health and safety practices for power related work	30	70	-	-	100	15
PSS/N1336.Working effectively with others	30	70	-	-	100	5
Total	60	140	-	-	200	20

Elective: 1 Erection and Stringing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N3501.Perform erection and stringing of a transmission tower	25	75	-	-	100	80
Total	25	75	-	-	100	80

Elective: 2 Foundation

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N3502.Construct the foundation of a transmission tower	30	70	-	-	100	80
Total	30	70	-	-	100	80



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
PPE	Personal Protective Equipment
KW	Kilowatt
V	Volt
KWH	Kilo Watt Hour
PTW	Permit To Work
CPR	Cardio Pulmonary Resuscitation
GIS	Geographical Information System

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.